

CHAPTER 3**CLIMATE****Learning Outcomes**

After studying this chapter the students will be able to:

1. Define climate.
2. Explain the reasons for differences in climate.
3. Differentiate between climate and weather.
4. Identify the general types of climate (based on latitude).
5. Identify the different climatic regions on a world map.
6. Compare different climatic regions.
7. Describe the climate in the different physical regions of Pakistan.
8. Describe how the climate of different physical regions affects the life over there.
9. Use given information to calculate the average temperature and monthly rainfall of different places.
10. Construct bar graphs from given climatic data.
11. Explain the various ways by which the human beings activities affect climate.
12. Explain how human beings activities are responsible for the greenhouse effect.
13. Identify individual and societal actions that can be taken to reduce adverse effects of human beings activities on climate.

CLIMATE

In this class we will learn about climate, but before we do let us recall what we learned about weather in class IV. In class IV, we learned that the Earth is surrounded by a layer of gases called the atmosphere. The change in the atmospheric conditions: temperature, humidity, atmospheric pressure, winds and precipitation (rainfall and snow), of a place at a specific time is known as weather. Climate is the average weather condition of a place. Let us learn how we come to know the climate of a place.

In order to describe the climate of a place the weather records, that is, the records of the weather conditions (temperature, rainfall) of that place over thirty years are studied. Look at figure 3.1, it shows the record of the average monthly temperature and the total monthly rainfall for December for Apple town over the period of 30 years.

Year	Average monthly temperature for December (°C)	Total monthly rainfall for December (mm)
1	25.5	3.0
2	25.3	3.1
3	25.5	3.2
4	25.4	3.4
5	25.3	3.3
6	25.8	3.5
7	24.3	3.1
8	25.3	3.6
9	24.5	3.4
10	25.3	3.8
11	25.6	3.9
12	24.3	4.5
13	24	5.5
14	23.3	5.4
15	23.7	5.3
16	23.9	5.1
17	23.5	5.1
18	22.2	5.3

19	21.2	4.6
20	22.3	3.2
21	21.6	4.2
22	21.5	4.2
23	21.5	5.4
24	21	3.2
25	21.3	3.4
26	21.4	3.5
27	21.6	6
28	21.7	5.8
29	21.8	5.3
30	21.9	4.2

Fig 3.1 Record of average monthly temperature and total monthly rainfall of Appletown for 30 years

In order to describe the climate of Appletown, meteorologists find the average weather experienced over 30 years as shown in figure 3.2

$$\text{3 years average temperature} = \frac{\text{Sum of average monthly temperature for 30 years for December (} ^\circ\text{C)}}{30}$$

$$\text{30 years average rainfall} = \frac{\text{Sum of total monthly rainfall for 30 years for December (mm)}}{30}$$

Fig 3.2 How to calculate climate of a place

Activity

- Using the information in figure 3.1 and the formula in figure 3.2 calculate the average temperature and monthly rainfall for December in Appletown.
- Use the data in figure 3.1 to prepare a bar graph showing the total monthly rainfall for Appletown.

The General Types of Climate in the World

No two places on the Earth have exactly the same climate. However, based on the latitude there are regions with similar average temperatures and average rainfall giving rise to four general types of climate:

- Equatorial climate.** This region lies between 5°N and 5°S of the Equator. The climate is generally hot. There is only one season, that is, summer.

- ii. **Tropical climate.** This region lies between 5° N and 23.5° N (Tropic of Cancer) and between 5° S and 23.5° S (Tropic of Capricorn). The climate is generally hot. There is a long and hot summer and a short and cool winter.
- iii. **Temperate (moderate) climate.** This region lies between 23.5° N (Tropic of Cancer) and 66.5° N (Arctic Circle) and between 23.5° S (Tropic of Capricorn) and 66.5° S (Antarctic Circle). The Climate here is moderate, that is, it is neither very cold in winter nor very hot in summer. This region enjoys all four seasons: summer, autumn, winter and spring.
- iv. **Polar climate.** This region lies between 66.5° N (Arctic Circle) and 90° N (North Pole) and between 66.5° S (Antarctic Circle) and 90° S (South Pole). The climate here is extremely cold with temperatures remaining below freezing point throughout the year. There is only one season, that is, winter.

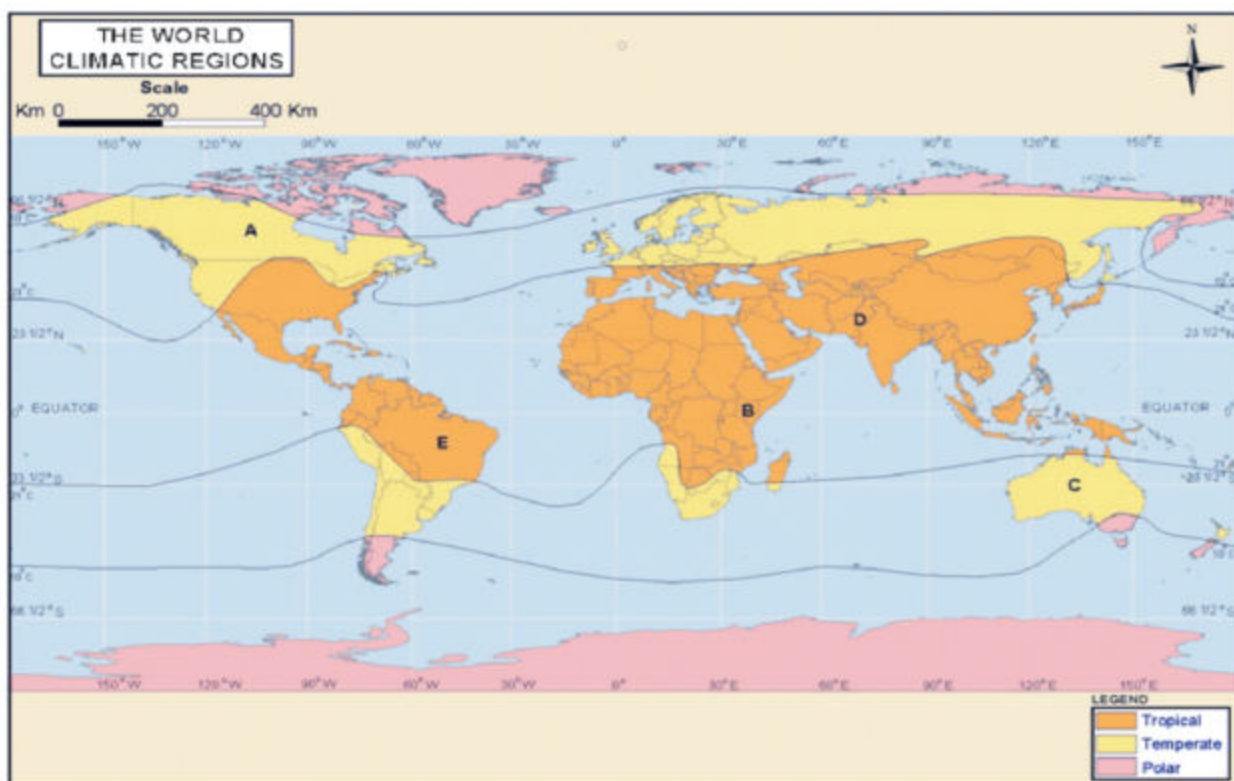


Fig 3.3 Major Climatic Regions of the World

In figure 3.3 we can see that the lines that help us identify the major climatic regions of the world are similar to the important lines of latitude in figure 1.6.

Some Reasons for the Difference in Climate

The difference in climate of countries, or in the many parts of a large country is caused by many factors. Here we will study how latitude or distance from the Equator, altitude (height above sea level) and distance from the sea, affect climate.

Latitude or distance from the Equator

Figure 3.3 shows that the hottest places are near the Equator. As we move from the equator towards the poles it becomes cooler. Figure 3.4 shows us that the temperature is higher at the Equator than at the poles because

- More of the sun's heat is hitting a smaller area of the Earth's surface at A as compared to B.
- The sun's rays have to travel a shorter distance. Therefore, less heat is lost during the journey to A than to B.

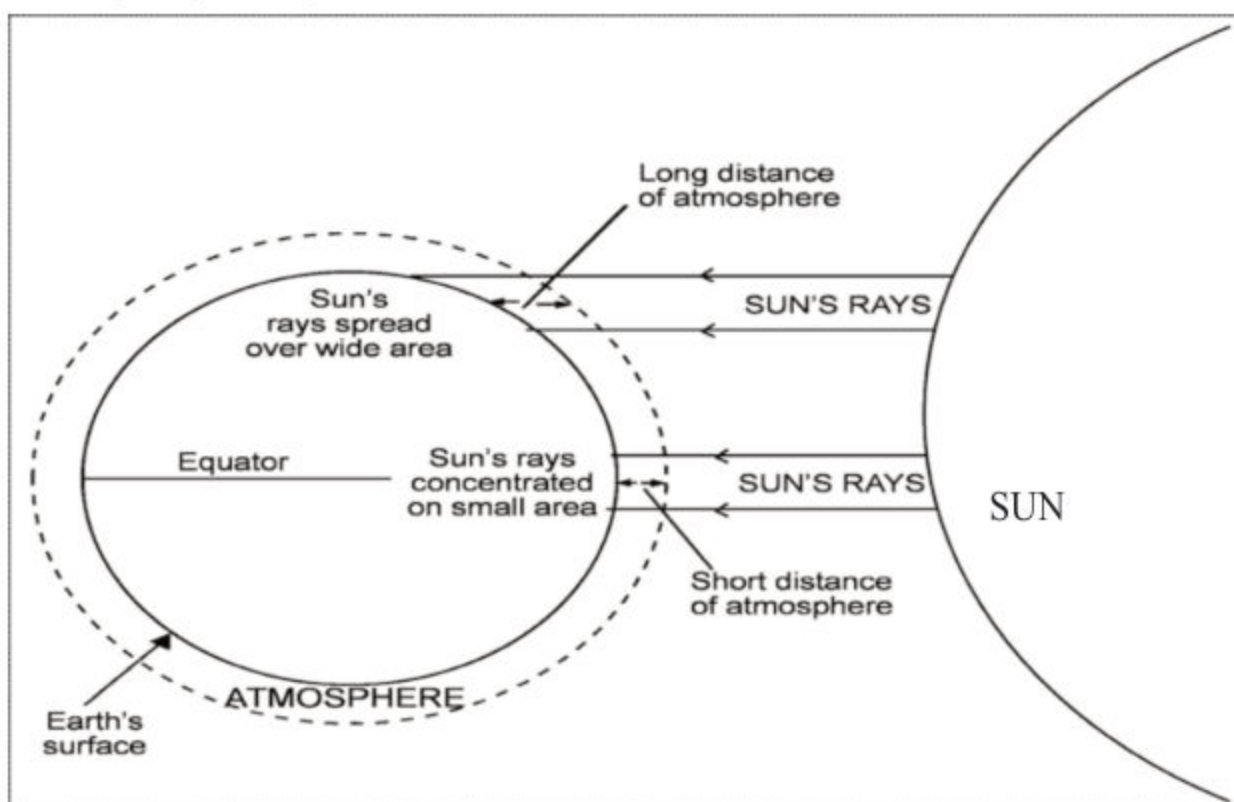


Fig 3.4 How latitude affects climate

Altitude or height above sea level

As we go higher, that is, more above sea level, the air becomes cooler at an average rate of 0.6°C for every 100 meters gained in height. So, it is cooler on the mountains than on the plains. For example, if the temperature on the plains (at sea level) is 30°C and the mountains nearby are at a height of 1000 meters then the temperature at the top of the mountains will be 24°C .

Distance from the Sea

As compared to the land, water heats up slowly and cools down slowly. Thus, the sea is cooler than the land in summer and warmer than the land in winter. During the summer, the sea is cooler than the land so places near the sea remain cool. In the winter, the sea is warmer than the land so places near the sea remain warm. As a result places which are near the sea do not have very big differences in temperature throughout the year. Places that are further inland are more likely to have extremes of temperature. For example, Karachi is near the sea and Jacobabad is farther away from the sea. Therefore, Karachi is cooler than Jacobabad in summer and warmer than Jacobabad in winter. Karachi also has less variation in its temperature throughout the year than Jacobabad does.

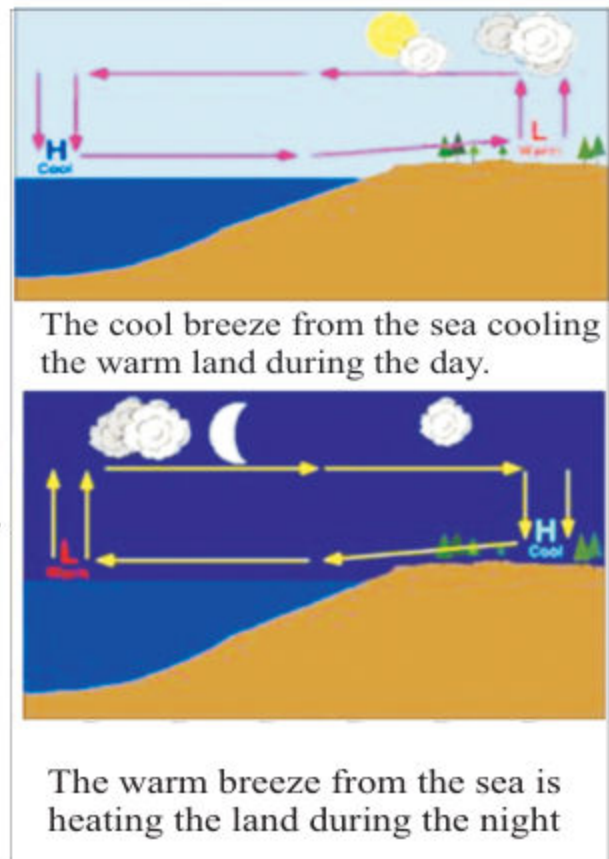


Fig. 3.5 How distance from the sea affects the climate

The Climate of Pakistan

From the map in figure 3.3 we can see that Pakistan lies just north of the Tropic of Cancer. It should, therefore, have a temperate climate. However, differences in



Fig 3.6 Map of Pakistan showing temperature at different places in January

latitude, altitude and distance from the sea means there are differences in the climate in different regions of Pakistan.



Fig 3.7 Map of Pakistan showing temperature at different places in June

In chapter 2 we learnt about the different physical regions of Pakistan. Let us see what the climate of each region is.

The Mountainous Regions

The north, north-east and north-west mountainous areas of Pakistan are high above sea level. Due to the high altitudes, winter here is cold and long. During winter, that is, October to April, the temperature stays below 10°C. Often times the temperature goes below freezing point (0°C). There is snowfall and the mountain peaks remain covered with snow throughout the year. In summer, that is, May to September, it is warm. June and July are the hottest months. However, the heat is often reduced by monsoon rain.

The Indus Plain Areas

The Punjab and most of Sindh are part of the Indus Plain areas. The summers are very hot and long. Summer extends from March to November and in northern Punjab from April to October. June is the hottest month. In the northern Punjab monsoon rains bring relief from the heat. In southern Punjab and Sindh, hot winds called “Loo” blow. Dust storms and thunderstorms occur during the summer. The thunderstorms cause heavy rainfall for a short time only. The rain causes the temperature to drop providing relief from the heat. Winter is mild with January

being the coolest month. The temperature in winter ranges from 10°C to 21°C .

The Coastal Areas

The narrow coastal strip of Pakistan experiences a long, warm summer and a short, mild winter. Summer is from March to November and winter is from December to February. Because of its closeness to the sea, in the summer the temperature in the coastal areas is lower than that in the Indus Plains and it is higher here than in the Indus Plains in the winter. During the winter season there are some cold days caused by cold winds blowing south from the Balochistan Plateau.

The Desert Regions

In the desert regions the summer is very hot and dry. Hot dusty winds blow throughout the summer. Winter is mild.

Activity:

Draw a table to show the temperature in each of the physical regions of Pakistan in January and June using the map in figure 3.6

How Climate Affects Us

The climate affects how we live. It affects such things as the clothes we wear, the food we eat, the sports we play, the work we do and the money we earn. Climate also affects where we live. Most people want to live in places that have a pleasant climate, that is, where it is neither extremely hot nor extremely cold. Therefore, most people in Pakistan live in the Indus plains.

i) Sports

Different climates offer different sports opportunities. For example, in cold, snowy climates, ice hockey, ice skating and skiing are popular sports. In warm and hot climates, people play sports such as cricket, hockey and football.



People skiing in the snow and playing field hockey

ii) **Dress**

The kind of clothing worn depends on the climatic conditions of a place. For example, thick garments are worn in places where the climate is cold so as to keep the body warm. In places where the climate is hot, thin garments are preferred, as they keep the body cool.



iii) **Food**

The people of cold regions usually eat meat, eggs, pulses and dry fruit. In warm and hot climates, the people like to eat vegetables, fresh fruits and curd and like to drink water, fruit juices and *lassi*.



iv) **Architecture**

Buildings are built to protect us from the heat and cold and to keep us comfortable. In cold climates, well insulated houses with slanting roofs are common. In hot climates, open and airy houses with straight roofs are common.



Fig. 3.8 How climate affects our lives

Effects of the Activities of Human Beings on Climate

With the development of human civilization, the impact of human activities on climate has become more obvious. For example, the world's average annual temperature is gradually rising which is due to rise in the amount of carbon dioxide gas in the atmosphere. Some of the reasons for the increase in carbon dioxide in the atmosphere are as follows:

- Human beings are rapidly cutting down forests. Plants take in carbon dioxide to prepare food. When forests are cut down it means more carbon dioxide remains in the atmosphere.
- The burning of fossil fuels (coal, oil, natural gas) in factories, vehicles, and homes releases large quantities of carbon dioxide into the atmosphere.
- Most of us like to eat beef. Cows release a large amount of methane into the atmosphere through belching and flatulence.



Fig. 3.9 Effects of human activities on climate

The Greenhouse Effect

The world is getting warmer. The Earth's atmosphere is warming up just as if it was trapped inside a greenhouse (see fig. 3.9). The warming effect is called the

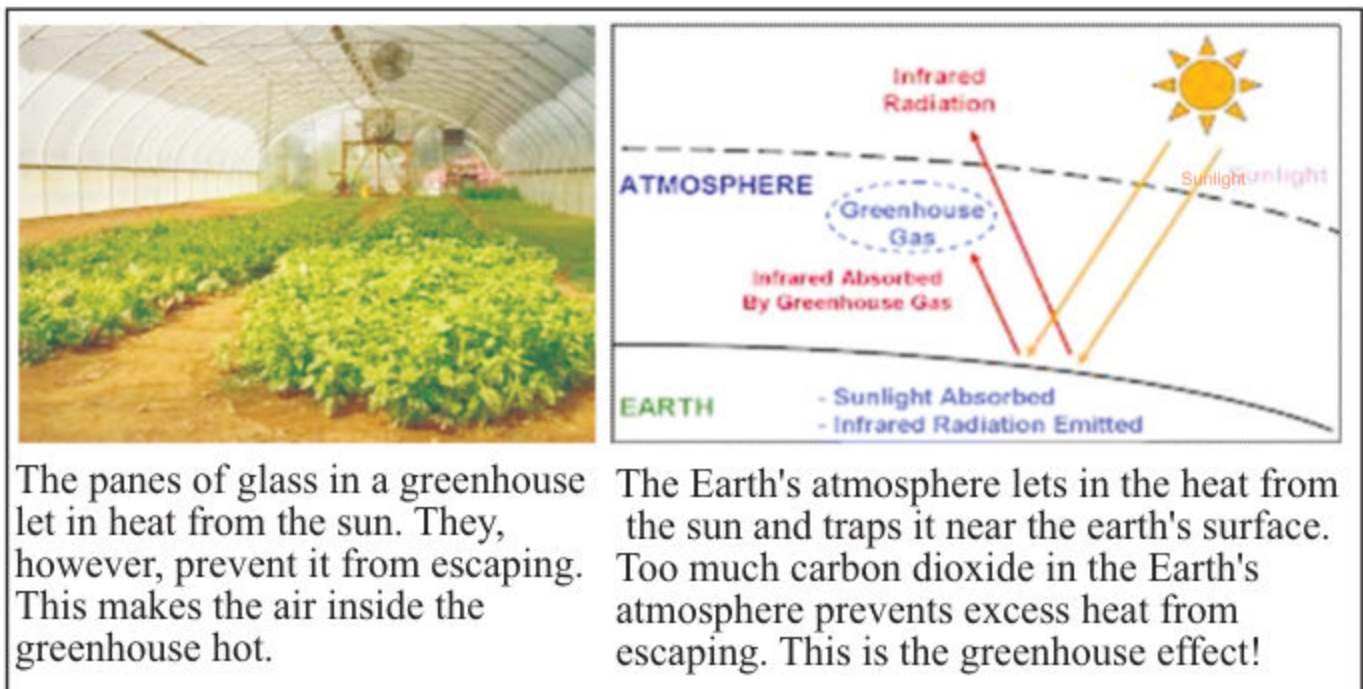


Fig. 3.10 The Greenhouse Effect

greenhouse effect. Why? In many places special houses are built to grow plants. These houses are called Greenhouses. The panes of glass or plastic sheets let heat from the sun into the green houses, but prevent it from escaping out. This makes the air inside the greenhouse hot.

Similarly, the Earth is warmed by the sun. The Earth's atmosphere lets in the heat from the sun absorbing what it needs near the Earth's surface and sending the rest of it back into the atmosphere. The carbon dioxide in the atmosphere traps some of this heat, stopping it from returning into space. This is necessary because it keeps the earth warm. But if there is too much carbon dioxide in the atmosphere, it prevents the excess heat from escaping, producing the same effect as the glass or plastic sheets do in greenhouses. This is the greenhouse effect and the gases, that is, carbon dioxide and methane that cause it are called Greenhouse Gases. As a result of the human activities discussed above Greenhouse Gases are produced in large quantities which cause the Greenhouse Effect and result in the rise of global temperature causing global warming.

Protection of the Environment

The air, climate, vegetation, landforms and living organisms collectively form what we call the environment. The protection of the environment from deterioration is essential for our survival. The following are a few actions which can help to protect the environment from the harmful impact of human activities:

1. Individual Actions

- i) We should practice the 3 Rs Reduce, Reuse and Recycle.
- ii) We should reduce our use of fossil fuels, like coal, petroleum and gas and use clean sources of energy like hydroelectricity, solar energy and wind energy.
- iii) We should reduce our use of wood so that we do not need to cut down forests and we should plant more trees.
- iv) We should reduce our use of energy by using saver bulbs to light our



homes and places of work, put the lights off when leaving a room and cycle or walk rather than using a motorbike or car if travelling only a short distance.

- v) We should reuse boxes and bottles and recycle paper, glass and plastic.
- vi) We should eat less beef and eat more vegetables and pulses.



Increasing use of solar energy

2. Societal Actions

- i) We can become part of a group working to protect the environment.
- ii) We should participate in efforts of civil society organizations to raise awareness of the need to protect the environment.
- iii) The government and civil society organizations should conduct reforestation programs regularly.
- iv) We should encourage the government to increase the generation of energy from environment-friendly sources.
- v) The public transport systems should be strengthened.
- vi) Household trash and other garbage should be properly disposed of to reduce the production of methane gas.



Public transportation

Fig. 3.11 Actions we can take to protect the environment

SUMMARY

In this chapter we have learnt what is climate, the difference between weather and climate, the general types of climate in the world and reasons for the differences in climate. We have also learnt about the climate in the different physical regions of Pakistan and the effects climate has on how we live our lives. Most importantly we have learnt how human activities are affecting climate and threatening our environment and what we can do both as individuals and as a society to protect the environment and save our planet Earth.

EXERCISE

A. Multiple choice questions

1. For every question, three answers are given; tick the correct one:
 - i. The region that receives vertical sunrays throughout the year is called
a. Equatorial Region b. Polar Region c. Tropical Region
 - ii. The climatic region that stretches between 23.5°N & 66.5°N and 23.5°S and 66.5°S latitudes is called
a. Equatorial Region b. Polar Region c. Tropical Region
 - iii. Which city of Sindh is located in the Coastlands?
a. Hyderabad b. Jacobabad c. Karachi
 - iv. Which gas do plants take in to produce their food?
a. Carbon dioxide b. Methane c. Oxygen
 - v. What is the average weather condition of a place called?
a. Climate b. Monsoon c. Weather

B. Answer the following questions

- i. Define the terms weather and climate.
- ii. Differentiate between weather and climate.
- iii. How do human activities affect climate?
- iv. What is meant by the greenhouse effect?
- v. List the positive and negative effects of climate.
- vi. List three individual actions and three societal actions that can have a positive effect on climate.
- vii. Explain the factors which affect climate.
- viii. List the general types of climate.

C. Complete the table

In the table below show the differences in the dress, food, recreation and housing because of the differences in climate.

Equatorial Climate	Tropical Climate	Temperate Climate	Polar Climate
Dress			
Food			
Recreation			
Houses			

D. Practical activities

1. Draw a map of Pakistan. Use different colours to show the different climatic regions on it.
2. On an outline map of the world, use different colours to show the general climatic regions on it.

Rain fall of Sindh 2001-2010

Year	Rainfall in (mm)
2001	129
2002	312
2003	97
2004	172
2005	65
2006	535
2007	199
2008	126
2009	231
2010	356

Ref: Pakistan Metrological Department Karachi, Pakistan.

3. Draw a bar graph to show in graphical form the 10 years of rainfall given above.
4. Look at the map of the world and figure 3.3. Name the areas marked A, B, C, D, E. Identify the type of climate: equatorial, tropical, temperate, polar for each named area.

E. Extension activities

1. Find out the climate of one city in each of the different climatic regions of Pakistan. Describe the life of people in that city.
2. Make a poster to raise awareness of the actions humans can take to protect the environment.